

Chakshu Gupta

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PROFESSIONAL SUMMARY

Computer Engineering student with strong foundations in **SQL, DBMS, Data Structures & Algorithms, and Python**. Experienced in **data analysis, problem-solving, and building AI/ML, LLM, RAG, and data-driven solutions** using Python, SQL, APIs, and Git. Hands-on experience integrating **AI tools, LLMs, and APIs** to develop and iterate on real-world applications. Solved **400+ algorithmic problems** on LeetCode.

EDUCATION

Thapar Institute of Engineering and Technology
Bachelor of Technology in Computer Engineering — CGPA: 8.40

Patiala, Punjab, India
Aug. 2023 – Present

WORK EXPERIENCE

Data Science Intern
Evoastra Ventures

Sept. 2025 – Nov. 2025
Remote

- Built a **Python** solution for **telecom customer churn analysis**, performing data ingestion, preprocessing, feature engineering, EDA, and analytical investigation across **10K+ customer records** to identify customer retention risks.
- Developed and evaluated a **Gradient Boosting Classifier** achieving **89% accuracy**; improved high-risk customer identification by **25%** through hyperparameter experimentation, cross-validation, and analysis of customer-level patterns.

RESEARCH EXPERIENCE

GAN-based Watermark Removal | *Research under Prof. Deep Maan*

Aug. 2025 – Present

- Implemented **Pix2Pix**, **Residual GAN**, and **Spatial Pyramid Attention** based **GAN models** for watermark removal across varying **watermark types**, with a **research paper submitted for peer review**.
- Achieved **44.29 dB PSNR** outperforming **baseline models** and demonstrated strong **generalization** across **watermark sizes, opacity variations, and image resolutions** under diverse **real-world degraded image conditions**.

PROJECTS

MediCure – AI Healthcare Platform | *FastAPI, PostgreSQL, LangGraph, LangChain* [🔗](#)*

Feb. 2026 – March 2026

- Built an end-to-end **AI healthcare platform** using **FastAPI, PostgreSQL, and LangGraph**, integrating patient monitoring, medical records, appointments, and **automated clinical alerts**.
- Engineered **multi-agent healthcare workflows** for symptom analysis, risk assessment, health checks, and **proactive health monitoring**, with persistent agent state, SSE-based responses, and real-time clinical alert generation.

Codebase Intelligence Assistant | *RAG, APIs, Vector Search, LLMs*

July 2026 – Aug. 2026

- Built an end-to-end **RAG solution** in **n8n** using the **GitHub API**, **Google Gemini embeddings**, and **Qdrant** to index repositories and enable semantic code search.
- Developed an automated **GitHub issue triage workflow** that retrieves relevant code through vector similarity search and uses an **LLM** to analyze root causes and generate suggested fixes, resolving **3072-dimensional vector** storage issues.

TrendyFire – Demand Forecasting System | *Gen AI, Machine Learning, NLP, BI* [🔗](#)*

Nov. 2025 – December 2025

- Built a **demand forecasting platform** integrating data collection, preprocessing, business analytics, and forecasting workflows.
- Automated **forecasting, reporting, and visualization** workflows through programmatic data processing, enabling faster analysis and iterative evaluation of demand trends.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, HTML, CSS
Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Algorithm Design, Problem Solving
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Transformers, Computer Vision, Feature Engineering, Data Preprocessing, Model Evaluation
APIs & Developer Tools: REST APIs, Git, GitHub, n8n, VS Code, Jupyter Notebook, Google Colab, Anaconda, Power BI
Libraries & Frameworks: PyTorch, TensorFlow, Scikit-Learn, pandas, NumPy, Matplotlib, OpenCV, SciPy, LangChain, LangGraph, FastAPI, Streamlit

CERTIFICATIONS

Generative AI: Introduction and Applications — IBM [Credential](#)
Generative AI: Foundation Models and Platforms — IBM [Credential](#)
NVIDIA: Fundamentals of Deep Learning — Whizlabs [Credential](#)